

Tiankuan (Felix) Wang

Master of Math (MMath) - Computer Science
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<https://felix-wang.top/>



Education

MMath Computer Science (Thesis)

University of Waterloo

Expected Graduation Date: 2026.12

Current GPA: 96/100

Relevant courses and skills:

- Advanced study of trustworthy machine learning, including adversarial machine learning, differential privacy (DP), model extraction and ownership, and backdoor attacks and defenses for generative AI systems.
- Advanced software engineering (SE) research, including empirical SE, software evolution, software analytics, mining software repositories, and data-driven SE research methodologies.
- Advanced study of AI for Software Engineering (AI4SE), focusing on LLM-driven code generation, testing, code review, software quality, software security, and AI-assisted software development.
- Formal verification techniques using proof assistants ([Lean 4](#), [Rocq](#)), focusing on compiler verification, domain-specific languages (DSLs), and formally verified software systems.

B.Sc. Computing Science (Honors)

University of Alberta

Graduated with First-Class Honors

Current GPA: 3.8/ 4.0

Relevant courses and skills:

- Cross-platform development using React Native, NestJS, and TypeScript, with a focus on Agile methodologies and software management practices.
- Database query implementation (SQLite, MongoDB) as well as database scaling and normalization.
- Software quality improvement using black-box testing, white-box testing, AB testing and performance testing.
- Statistical natural language processing (NLP), including Hidden Markov Models (HMMs), probabilistic context-free grammar (PCFG) parsing, Naïve Bayes, and hands-on implementation of classical NLP algorithms.
- Machine learning fundamentals, including convex optimization, empirical risk minimization (ERM), regression, classification, and neural networks, with hands-on implementation of machine learning algorithms.

Work Experience

Teaching Assistant (TA)

2025.9 - current

University of Waterloo – Waterloo, Canada

- CS 136 – Elementary Algorithm Design and Data Abstraction (Fall 2025, Spring 2026).
- CS 431 – Data Intensive Distributed Computing (Winter 2026).

Software Engineer Intern

2023.7 - 2023.8

Haier Smart Home - Qingdao, China

- Participated in the development of the speech control module for the Casarte (Haier) washing machine.
- Contributed to the code of Haier's IoT packages, including the startup sound feature, voice-switching feature, sleep mode voice activation feature, and washing and drying voice activation feature.
- Participated in group meetings to report project details and coordinate project progress with project managers.

Research Assistant (RA)

2022.5 - 2022.9

Dalhousie University Software Engineering Lab - Halifax, Canada

- Worked on an interview study discussing the Software Industry's hybrid working mode during the Covid pandemic.
- Supervised by award-winning computer scientist Dr. Paul Ralph.
- Conducted interviews, collected data, and transcribed the information for analysis.
- Presented research findings to groups of people.
- Performed qualitative analysis on interview data.
- Engaged in academic writing to contribute to scholarly discourse.

Research Experience

- **SecVecCoder Model:** Mentored by Dr. Mei Nagappan and Dr. N. Asokan on post-training alignment of code Large Language Models (LLMs) using task-vector arithmetic to jointly improve code functionality and security.
- **Traffic Prediction Neural Networks:** Mentored by Dr. Zesheng Cheng on traffic flow prediction projects, contributing to research on unsupervised graph representation learning and SVD-enhanced STGCN models.
- **Interview Study:** Mentored by Dr. Paul Ralph on an interview study about the hybrid working mode in Hybrid Software Engineering during COVID. Compared teams under hybrid working mode, co-located software engineering mode and fully remote working mode.

Course Projects

Smart Home (TypeScript, React Native, NestJs, PostgreSQL, Docker, Nginx)

- A TypeScript project by a startup company aiming to manage home appliances and provide maintenance solutions. Use React Native as a presentation layer, Nginx and NestJs for business and persistence, and PostgreSQL for DB service. The project is Dockerized and deployed on Cybera Cloud.
- GitHub Page: <https://github.com/UAlberta-CMPUT401/f23project-simply-home>

Formally Verified LLM Tokenizer (Lean 4)

- A formally verified Byte Pair Encoding (BPE) tokenizer for LLMs in Lean 4, guaranteeing that the internal token representation is a lossless representation of the original text.
- GitHub Page: <https://github.com/U70-TK/BPETokenizer>
- Report: <https://felix-wang.top/LeanBPE.pdf>

LLM-Assisted Software Security Analysis

- Developed an LLM-assisted pipeline to identify and classify real-world pull requests into software vulnerability categories, enabling empirical analysis of 933k pull requests authored by AI coding agents and human developers.
- GitHub Page: <https://github.com/U70-TK/MSR-2026-challenge>
- Report: https://felix-wang.top/AIDev_report.pdf

AI Pair Programming User Study

- Designed a novel human–AI interaction framework through a controlled user study (N = 8), investigating how multiple LLM-generated code solutions influence developer decision-making and software security.
- GitHub Page: <https://github.com/U70-TK/MSR-2026-challenge>
- Report: https://felix-wang.top/user_study.pdf

Skills

Primary Research Area:

- Software Engineering (AI for Software Engineering, AI4SE).
- Large Language Models (LLMs) and AI Alignment.
- Trustworthy code generation.
- Human Factors in Software Engineering.

Technical Skills:

- **Programming Languages:** Python, C/C++, Java, TypeScript, JavaScript, HTML/CSS, Lean 4.
- **Frameworks & Tools:** Git, Docker, React, NestJS, Apache Spark.
- **Databases:** PostgreSQL, MySQL, MongoDB, Firebase.
- **Machine Learning:** PyTorch, Hugging Face ([Transformers](#), [TRL](#), [PEFT](#)), [vLLM](#), [Axolotl](#).

Awards and Scholarships

- 2024 Dean's Honor Roll – University of Alberta Faculty of Science.
- International Master's Award of Excellence (IMAE) – University of Waterloo (CAD \$2500 per term).
- Graduate Research Studentship (GRS) – University of Waterloo (CAD \$6818 per term).